What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Hodgkin Lymphoma

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Adapted from: Hodgkin Lymphoma

Listen to this Brick
0:00 / 22:20 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of Hodgkin lymphoma (HL) from other lymphomas.
2. Explain the pathophysiology and five histologic subtypes of HL.
3. Interpret clinical findings to properly diagnose HL.
4. Outline the

CASE CONNECTION

JM is a 26-year-old male who comes to your clinic because he's been experiencing weight loss and night sweats. "I've always been healthy, but now I'm worried. About 3 months ago, I noticed that my clothes were getting looser, but I'm not trying to lose weight. I'm also waking up in the middle of the night all drenched with sweat." JM's vital signs are within normal limits; however, his record documents a 9-kg (20-lb) weight loss since his last visit 6 months ago. On physical examination, multiple small, nontender cervical lymph nodes are palpated in his neck, axilla, and groin.

How will you discuss work-up and treatment options with JM? What is his prognosis? Consider your answers as you read, and we'll revisit JM at the end of the brick.

GO TO CONCLUSION ↓

What Is Hodgkin Lymphoma?

Hodgkin lymphoma (HL) is a type of hematopoietic lymphoid neoplasm characterized by scattered large atypical malignant cells (Reed-Sternberg cells) in a background of benign lymphocytes/inflammatory cells.

Reed–Sternberg Cell

Hodgkin lymphoma was first described in 1832 by English physician Thomas Hodgkin, but the mal , the Reed-Sternberg (RS) cell, remained unknown for many years. In fact, the original name "Hodgkin disease" was devised because, at the time, it was not clear that this was a lymphoma.

Epidemiology

HL is a rare cancer that affects fewer than 10,000 people in the United States each year, and the average age of diagnosed patients is 35 years. However, if you look closer, there are two peak age groups: it is one of the most common cancers of adolescents and young adults, and it peaks again in patients older than 65 years (Figure 1).

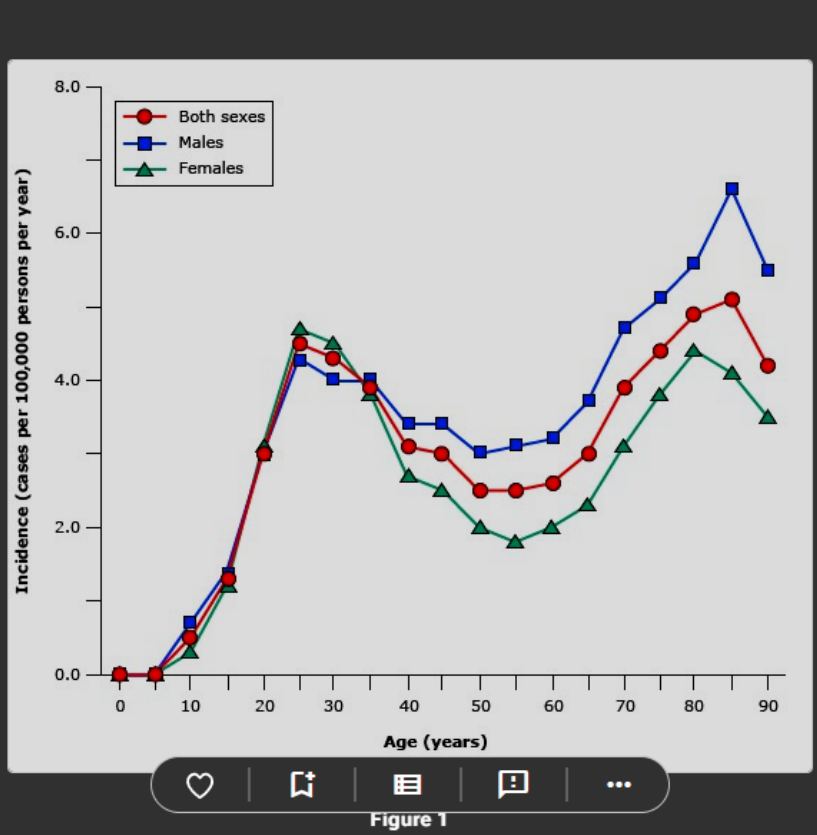


Figure 1

Which two demographic groups are most at risk for Hodgkin lymphoma (HL)?

How Do Patients With Hodgkin Lymphoma Present (LO #1)?

Like many lymphomas, HL has a variable, and often vague, presentation. Many patients are asymptomatic at diagnosis, and the disease is found when the clinician feels adenopathy (enlarged lymph nodes) or sees a mediastinal mass on a chest x-ray.

Approximately one-third of patients may have the so-called **B symptoms** seen in many lymphomas: fevers, night sweats, and weight loss. Patients may also experience fatigue.

Nearly half of the patients have evidence of mediastinal lymphadenopathy on chest x-ray at diagnosis. A mediastinal mass that compresses surrounding structures such as the superior vena cava and airway may cause dyspnea and facial swelling.

How does Hodgkin lymphoma (HL) typically present?

CLINICAL CORRELATION

In contrast to non-Hodgkin lymphoma, Hodgkin lymphoma typically demonstrates orderly, contiguous lymph node spread, involves more central and axial lymph nodes (seldom involving mesenteric nodes and Waldeyer's ring), and rarely has an extranodal presentation.

What Is the Pathophysiology of Hodgkin Lymphoma (LO #2)?

In this section, we will look at the risk factors for HL, histologic appearance of RS cells, histologic subtypes, and complications.

Risk Factors

Epstein-Barr virus (EBV) infection has been identified in as many as 40% of HL cases. Epstein-Barr DNA has been isolated from some malignant cells.

Human immunodeficiency virus (HIV) infection is also associated with HL; most of these patients are also EBV+. Cigarette smoking, obesity, and immunosuppressive disorders are additional risk factors.

Pathology

As mentioned earlier, the **Reed-Sternberg** cell is the most distinct morphologic feature of Hodgkin lymphoma (Figure 2).

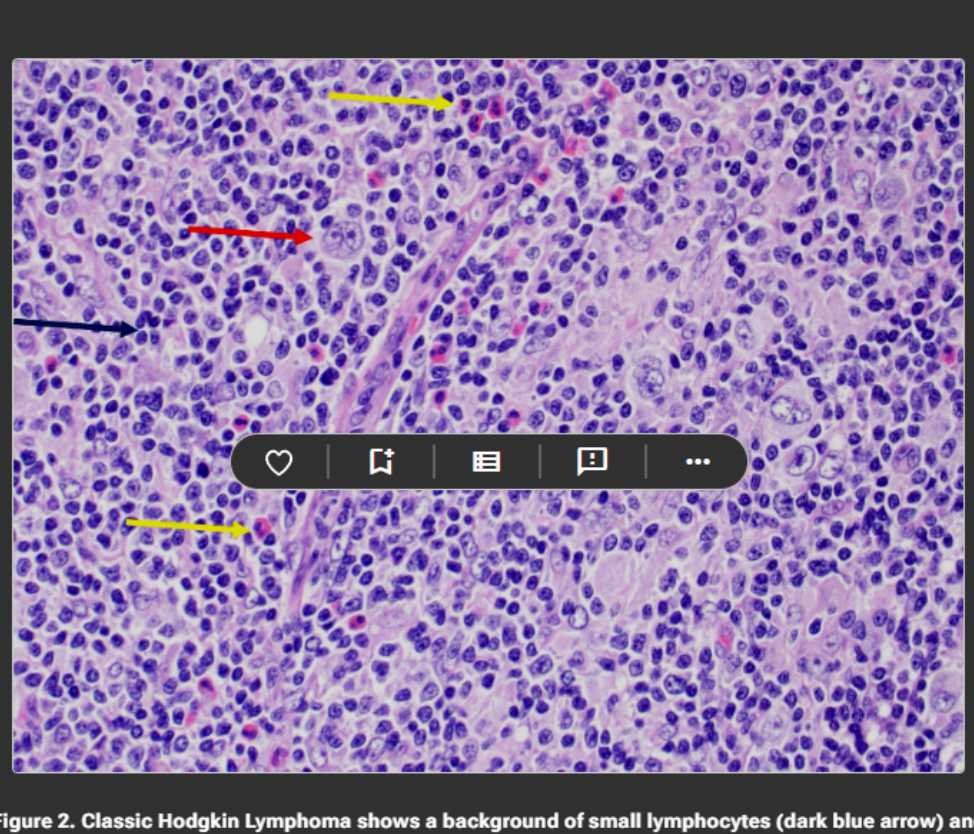


Figure 2. Classic Hodgkin Lymphoma shows a background of small lymphocytes (dark blue arrow) and eosinophils (yellow arrows) with scattered Reed-Sternberg cells (red arrow). RS cells are binucleated with centrally located nucleoli, giving them an "owl's eye" appearance.

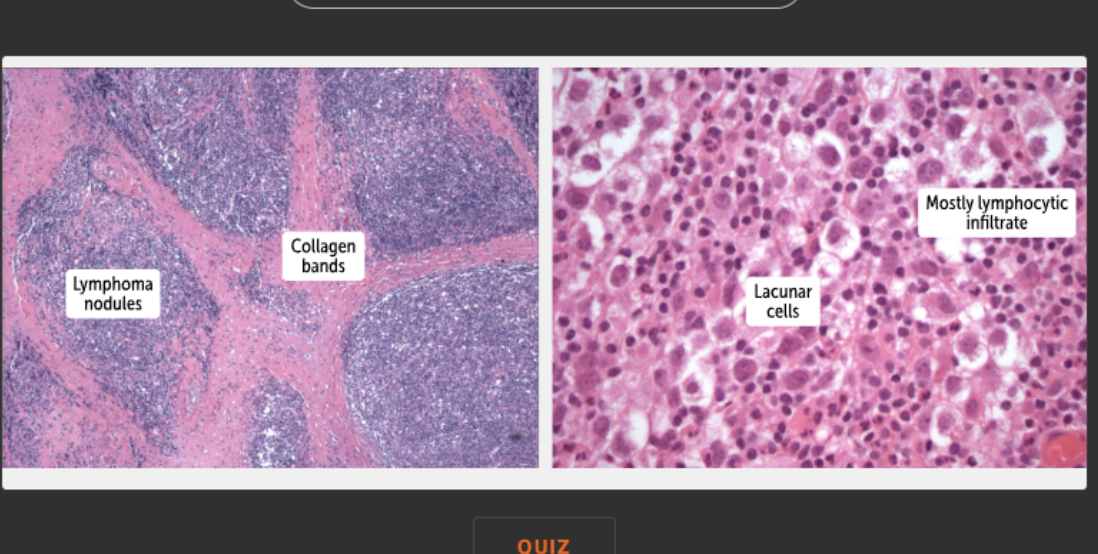
Reed-Sternberg Cell Appearance. Classic RS cells are binucleated with centrally located nucleoli, giving them an "owl's eye" appearance. Benign reactive lymphocytes are often found, and these can make up the bulk of the tumor.

Reed-Sternberg Immunotyping. Reed-Sternberg cells stem from germinal or post-germinal B cells with defective Ig genes, so they have lost many of their B-cell markers. Reed-Sternberg cells are negative for CD45 and CD20 but positive for CD15 and CD30.

Histological Subtypes

Nodular Sclerosis. This is the **most common** type of HL in the United States (approximately 70-75% of cases), and its prognosis is excellent because it most commonly occurs in young adults and typically presents at an early stage.

In this subtype, fibrous bands divide the lymph nodes into nodules as seen in Figure 3, left.



QUIZ

Figure 3

This subtype may have RS cell variants called "lacunar" cells, which are giant cells surrounded by clear space (Figure 3, right). They may also have the classic RS cells (not seen in this figure).

Mixed Cellularity. The second most-common type of HL in the United States is mixed cellularity, accounting for approximately 20% of cases. The mixed cellularity type contains classic RS cells and a reactive inflammatory infiltrate but lacks fibrous bands. The mixed cellularity subtype tends to occur in the older population and can present with more advanced disease. This subtype is more often associated with EBV.

Less Common Subtypes. The other two less-common types of classic HL are **lymphocyte-rich HL**, which has more reactive mature lymphocytes than classic RS cells, and l abundant classic RS cells but few reactive lymphocytes (more often associated with EBV and HIV infection).

Nodular Lymphocyte-Predominant. This is an uncommon subtype (~5% of all Hodgkin lymphomas) that has historically been grouped as a "Hodgkin lymphoma" because of their similar morphologies; however, it has an opposite immunophenotype, different genetics, no EBV association, and no bimodal age distribution. Therefore, newer classifications are labeling this tumor as "nodular lymphocyte-predominant B-cell lymphoma".

This lymphoma occurs more often in young males, presenting with localized disease. Cervical, axillary, or inguinal lymph nodes are most commonly involved. Morphology demonstrates a background of small lymphocytes and occasional large cells with convoluted nuclei, known as "popcorn" cells. These large cells, in contrast to Reed-Sternberg cells, are positive for CD45 and CD20 and negative for CD15 and CD30.

These patients typically experience an indolent clinical course with long survival; however, they may transform into a higher-grade lymphoma (such as T-cell/histiocyte-rich large B-cell lymphoma).

How Do We Diagnose Hodgkin Lymphoma (LO #3)?

The most common physical exam finding in HL is lymphadenopathy (enlarged lymph nodes), characterized by multiple large, hard, fixed lymph nodes in one or more body regions.

In addition, patients with massive mediastinal lymphadenopathy may have signs of superior vena cava (SVC) syndrome, in which SVC compression leads to facial erythema; faci veins in the face, upper chest, and upper limbs; and headache. Airway compression can cause respiratory distress and stridor.

Occasionally, HL is diagnosed when a patient with weight loss or fever receives a chest x-ray that shows a mediastinal mass.

Lymph Node Biopsy

In patients who have lymphadenopathy without an underlying infection or inflammatory disorder, we next do a lymph node biopsy. This provides a definitive diagnosis of HL using the histologic characteristics described above.

After the tissue diagnosis is made, we proceed to staging the lymphoma, which will guide treatment. This is done using physical exam findings, blood tests, and radiology.

Radiologic Imaging. Radiologic testing is crucial to stage HL. The extent of the lymphoma and its effect on surrounding organs is determined by positron emission tomography (PET)/CT scanning. Evidence of a mass compressing the right heart and displacing the superior vena cava is indicated by the arrow in Figure 4.

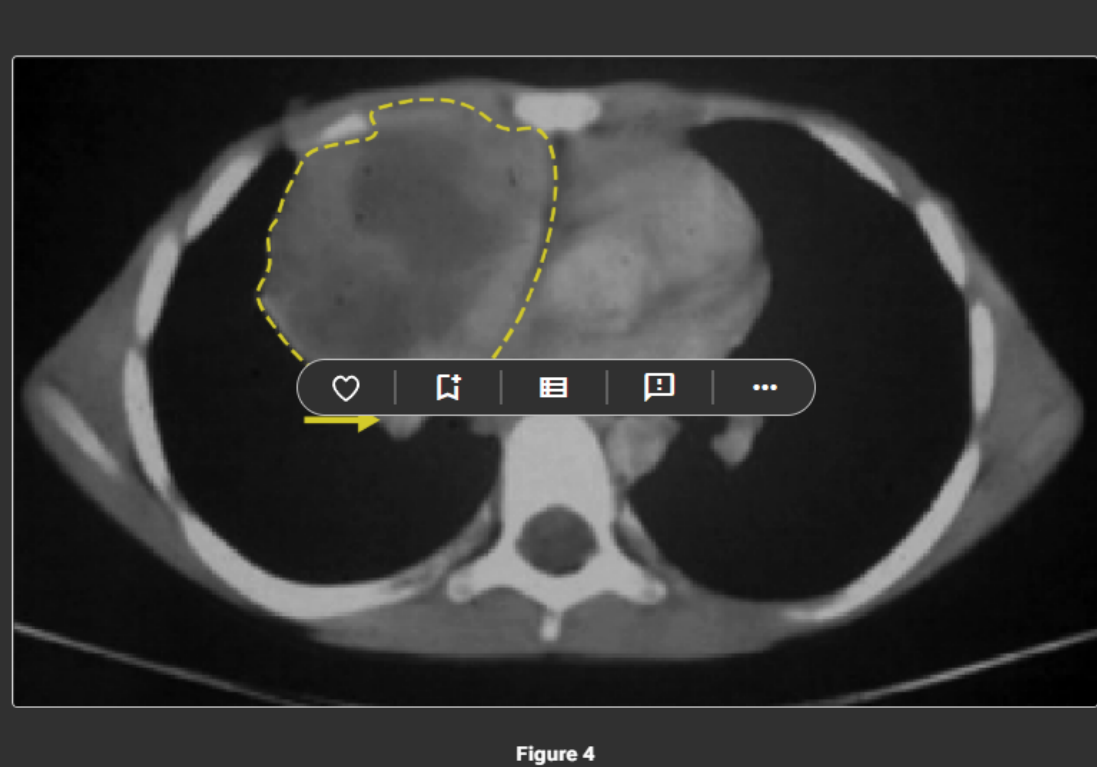


Figure 4

An F-18 fluorodeoxyglucose (FDG) PET scan is useful for staging HL. Figure 5 shows a patient with multiple lymph node chain involvement above and below the diaphragm.

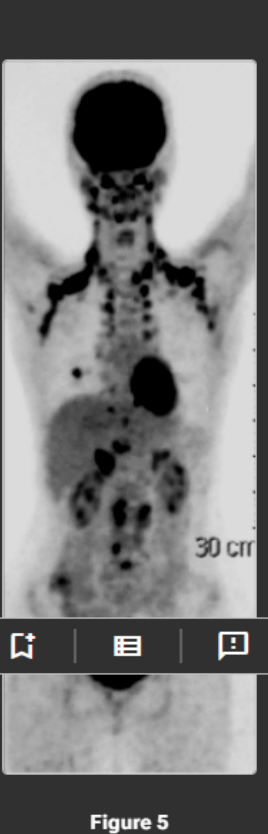


Figure 5

Laboratory Analysis. During staging, infiltration of the bone marrow is suggested by pancytopenia on complete blood count (CBC). Increased blood cell turnover (typical of some HL cells) is suggested by an elevated serum potassium, lactate dehydrogenase (LDH), or erythrocyte sedimentation rate (ESR).

Staging of Hodgkin Lymphoma

Staging HL considers factors such as the number and location of affected nodes, liver or bone involvement, and presence of B symptoms.

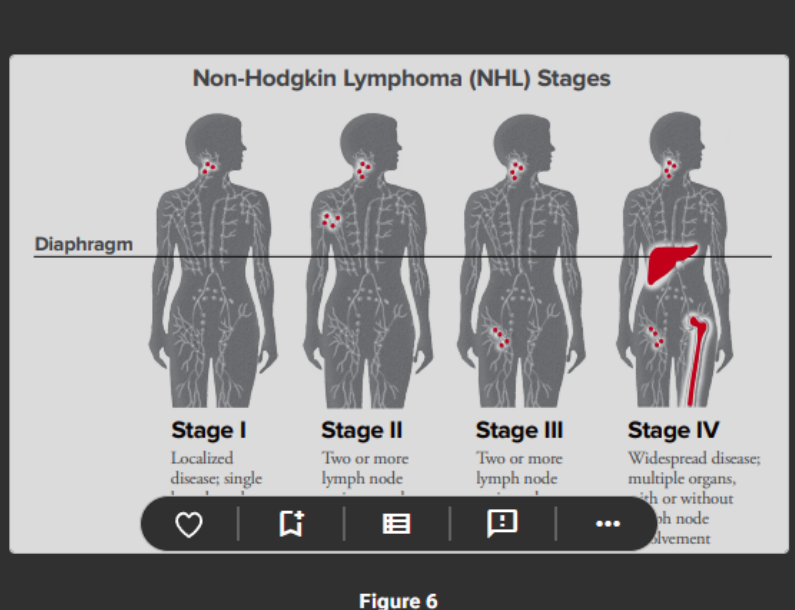

Non-Hodgkin Lymphoma (NHL) Stages
Diaphragm
Stage I
Localized disease; single
Stage II
Two or more lymph node
Stage III
Two or more lymph node
Stage IV
Widespread disease; multiple organs, with or without lymph node involvement

Figure 6

For Hodgkin lymphoma, this staging is modified depending on the presence or absence of B-symptoms. For example, stage II without B symptoms is stage IIA, and stage II with B symptoms is stage IIB.

At which stage of Hodgkin lymphoma (HL) is lymph node involvement first seen on both sides of the diaphragm?

How Do We Manage Patients With Hodgkin Lymphoma (LO #4)?

Treatment for the early stages of HL consists of radiation therapy (RT) and chemotherapy. This combination has a higher disease-free survival rate. The radiation is site or fie ion controls growth at the known tumor, and chemotherapy controls any occult disease outside the radiation field.

A common chemotherapy regimen used in HL treatment **is ABVD** (Doxorubicin, Bleomycin, Vinblastine, Dacarbazine). (Note: The A in ABVD is Adriamycin, the brand name for Doxorubicin)

ABVD is administered every 14 days for 28 days.

Doxorubicin (Adriamycin): Mechanism: Inhibition of DNA/RNA synthesis. Cumulative dose-related cardiomyopathy can occur, which can also cause heart failure.

Vinblastine: Mechanism: Inhibition of microtubule function. Dose-related peripheral neuropathy is associated with use, which can cause paresthesia, loss of reflexes, weakness, and autonomic paralysis (including vocal cord paralysis)

Bleomycin: A metal-chelating, chemotherapy antibiotic that has a cytotoxic effect on nondividing tumor cells by causing fragmentation of DNA chains. Adverse effects include fever, hypersensitivity reactions, skin changes, and severe pulmonary fibrosis.

Dacarbazine: This non-cell cycle-specific purine analog inhibits DNA polymerase and leads to DNA strand breaks. Adverse effects: Nausea and Vomiting (>90%), bone marrow suppression, leukopenia, and thrombocytopenia are common.

In cases of refractory or relapsed HL, autologous stem cell transplantation is used. In addition, salvage therapy regimens include targeted immunotherapy agents such as brentuximab or pembrolizumab.

Brentuximab is a mo CD30 cell surface receptors that are hig phoma RS cells.

Pembrolizumab: A monoclonal antibody and immune checkpoint inhibitor targeting programmed cell death-1 (PD-1), used to treat unresectable or metastatic solid tumors with genetic anomalies like mismatch repair deficiency and microsatellite instability.

Prognosis

The stage of the lymphoma largely determines prognosis.

Prognosis is good in patients with early stage disease who may achieve complete remission. Disease-free survival at 5 years ranges from 7% to 42% in patients with advanced disease.

What is the standard treatment regimen for patients with Hodgkin lymphoma (HL)?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JM, how could you discuss his work-up and treatment options? What is his prognosis?

After a history and physical examination, the presumptive diagnosis is lymphoma. A cervical lymph node biopsy reveals Reed-Sternberg cells, diagnostic for Hodgkin lymphoma. To stage the disease and determine prognosis and treatment, a positron emission tomography (PET)/CT scan of the chest/abdomen/pelvis is performed. Multiple enlarged lymph nodes are detected above the diaphragm as well as a small anterior mediastinal mass.

JM can now be told that he has early stage Hodgkin lymphoma, but his prognosis is good, as it typically is for patients in the early stages who receive standard treatment. JM is referred to a hematologist-oncologist. When he asks about treatment, you tell him it is likely he will receive multiple chemotherapeutic agents and radiation treatment to the involved lymph nodes. "But let's see what the specialist says," you say.

Summary

- Hodgkin lymphoma (HL) is a rare hematopoietic lymphoid neoplasm characterized by Reed-Sternberg (RS) cells embedded in an abundant benign lymphocytes/inflammatory cells.
- HL typically presents with enlarged lymph nodes or a mediastinal mass on a chest x-ray, and sometimes with weight loss, night sweats, and fever.
- Risk factors for HL include Epstein-Barr virus (EBV) infection, human immunodeficiency virus (HIV) infection, cigarette smoking, obesity, and immunosuppress
- Histologic subtypes of HL include nodular sclerosis, mixed cellularity, lymphocyte-rich, and lymphocyte-depleted.
- Nodular lymphocyte predominant B-cell lymphoma has a similar morphology but is now not considered a subtype of HL, in part due to its different immunophenotype (popcorn cells positive for CD45 and CD20 and negative for CD15 and CD30).
- HL is diagnosed through a lymph node biopsy and staged using physical exam findings, blood tests, and radiologic imaging.
- Treatment for HL usually involves chemotherapy, with or without targeted radiation therapy.
- The prognosis for HL largely depends on the stage of the lymphoma, with a good prognosis for patients with early stage disease.

Review Questions

1. A 40-year-old male presents with intermittent fever and 6.8-kg (15-lb) weight loss over 3 months. His vital signs are normal. Multiple 3 cm, nontender, hard, mobile cervical nodes are present. The rest of the exam is normal. Which of the following is the most appropriate next diagnostic step in this patient?

A. ● Cervical node biopsy
B. ○ Chest and abdominal CT scan
C. ○ Chest x-ray
D. ○ Erythrocy

Hide Explanation

The correct answer is cervical node biopsy (A). This patient with hard cervical nodes, weight loss, fever, and fatigue most likely has lymphoma. Biopsy of accessible lymph nodes, such as cervical nodes in this case, will provide tissue diagnosis of lymphoma. After histologic confirmation, chest and abdominal CT (B) will help us stage lymphoma but it will not provide an initial diagnosis. Chest x-ray (C) might show anterior mediastinal mass but cannot make an initial diagnosis. An elevated erythrocyte sedimentation rate (D) is used in staging some lymphomas like Hodgkin lymphoma, but would not be the initial test done.

2. Infection with which of the following microorganisms is most strongly associated with Hodgkin lymphoma?

A. ● Epstein-Barr virus
B. ○ *Helicobacter pylori*
C. ○ Human herpesvirus-8
D. ○ Human papillomavirus
E. ○ *Schistosomiasis haematobium*

Hide Explanation

The correct answer is Epstein-Barr virus (A). Its DNA has been found in Reed-Sternberg cells in approximately 40% of cases. However, it remains unclear whether this virus plays a pathogenic role. *Helicobacter pylori* (B) has been linked to MALT (mucosa-associated lymphoid tissue) lymphoma and is associated with gastric ulcers. The oncogenic human papillomavirus (D), particularly HPV16 and HPV18, has been linked to cervical and oropharyngeal cancers. Human herpesvirus-8 (C) may cause Kaposi sarcoma in patients with AIDs. *Schistosomiasis haematobium* (E) chronically irritates the bladder epithelium, causing squamous cell carcinoma of the bladder.

3. Histologic exam of cervical lymph nodes from a patient with diffuse lymphadenopathy shows a nodular growth pattern with few CD30-negative mononuclear giant cells. Which of the following is the most likely Hodgkin lymphoma subtype in this patient?

A. ○ Lymphocyte-depleted
B. ○ Lymphocyte-rich
C. ○ Mixed-cellularity
D. ● Nodular lymphocyte-predominant
E. ○ Nodular sclerosis

Hide Explanation

The correct answer is nodular lymphocyte-predominant (D). This patient with generalized lymphadenopathy, nodular growth pattern, and CD30-negative mononuclear giant cells suggestive of nonclassical Reed-Sternberg (RS) cells most likely has nodular lymphocyte-predominant Hodgkin lymphoma (HL), though it is now more appropriately termed nodular lymphocyte-predominant B-cell lymphoma. Lymphocyte-depleted (A) HL has abundant classic RS cells (CD30 and 15 positive) with few reactive lymphocytes. Lymphocyte-rich (B) HL has few classic RS cells with abundant reactive lymphocytes. Mixed-cellularity (D) HL has few classic RS cells with abundant inflammatory cells including neutrophils, lymphocytes, basophils, eosinophils, and macrophages. Nodular sclerosis (E) HL has classic RS cells and nodular growth pattern due to bridging fibrosis.

4. Which drug used to treat Hodgkin lymphoma is most likely to cause severe pulmonary fibrosis?

A. ● Bleomycin
B. ○ Dacarbazine
C. ○ Doxorubicin
D. ○ Vinblastine

Hide Explanation

The correct answer is Bleomycin (A). Bleomycin is known for causing severe pulmonary fibrosis a ciated with nausea, vomiting, and bone marrow suppression, but not pulmonary fibrosis. Doxorubicin (C) is associated with cardiomyopathy. Vinblastine (D) is associated with peripheral neuropathy.

How would you rate this Brick?

References

- Gorczyca W. *Atlas of Differential Diagnosis in Neoplastic Hematopathology*. 4th ed. CRC Press; 2021.
- Greer JP, et al, eds. *Wintrobe's Clinical Hematology.* 14th ed. Wolters Kluwer; 2018.
- https://acsjournals.onlinelibrary.wiley.com/doi/10.3322/caac.21438
- https://www.nature.com/articles/s41572-020-0189-6
- https://www.uptodate.com/contents/hodgkin-lymphoma-epidemiology-and-risk-factors?search=hodgkin%20lymphoma&source=search_result&selectedTitle=10~150&usage_type=default&display_rank=11
- Kumar V, Abbas AK, Aster JC. Hematopoietic and Lymphoid Systems. In: Kumar d Cotran *Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.
- https://www.uptodate.com/contents/hodgkin-lymphoma-epidemiology-and-risk-factors?search=hodgkin%20lymphoma&source=search_result&selectedTitle=10~150&usage_type=default&display_rank=11
- Kumar V, Abbas AK, Aster JC. Hematopoietic and Lymphoid Systems. In: Kumar V, Abas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.
- Provan D, ed. *ABC of clinical haematology.* John Wiley & Sons, Incorporated; 2018.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of Hodgkin lymphoma (HL) from other lymphomas.~~
02 ☑ ~~Explain the pathophysiology and five histologic subtypes of HL.~~
03 ☑ ~~Interpret clinical findings to properly diagnose HL.~~
04 ☑ ~~Outline the management plan of HL.~~

Objective Confidence: 4/4